

Concept paper (proposal) for the Semester Thesis

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(Project-) Title: Hybrid Modelling of Anisotropy using CNN

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Name of supervisor:

1. Topic

Anisotropy refers to the property of a material that exhibits different mechanical behaviours when subjected to loads in different directions. Modelling anisotropy mathematically is challenging due to varying microstructure of the crystal or grain, nonlinear mechanical behaviour of the material. In order to accurately model anisotropy, it is often necessary to develop models that span multiple length and time scales. This can be difficult due to the need to incorporate different physical mechanisms and the potential for computational complexity.

To tackle the above-mentioned problems, this project focuses on training neural networks to predict the mechanical properties of anisotropic materials based on grain orientations and deformations as the input parameters.

2. State of science and technology

The use of neural networks to model anisotropy in multiscale models is an area of active research in materials science and engineering. Multiscale models are important for accurately capturing the behaviour of anisotropic materials, as they can account for the complex microstructures and physical mechanisms that give rise to anisotropy.

One approach to using neural networks in multiscale modelling is to develop hybrid models that combine atomistic or molecular dynamics simulations with continuum models. Researchers have used datasets obtained from molecular dynamics simulations to train ANNs that can approximate unambiguous state functions, such as the elastic stress–strain relation (tensor equation of state), the dislocation nucleation threshold, the void nucleation threshold and the beginning of bcc transformation. (Alexander E. Mayer, 2022).

Another approach is to use neural networks to develop surrogate models that can replace more computationally expensive models, such as finite element models. GNN Surrogate models can be trained to predict the grain-scale elastic response of polycrystalline metallic alloys and can provide accurate predictions at a fraction of the computational cost of traditional models. (Darren C. Pagan, 2022)

Additionally, researchers are exploring the use of neural networks to develop data-driven models that can capture the anisotropic behaviour of materials at different length scales. For example, researchers have used neural networks to predict the mechanical properties of polycrystalline materials based on the crystallographic orientations of the individual grains, and have shown that these predictions can be used to develop accurate multiscale models. (Minyi Dai, 2021)

Overall, the use of neural networks in multiscale modelling of anisotropy is a rapidly developing area of research, and has the potential to greatly improve our understanding of the behaviour of anisotropic materials in various applications.

3. Objective of your project

To proceed with the scientific project, extensive information about predicting mechanical behaviour using convolutional neural networks would be gathered. This would include a review of existing literature, data sets, and relevant research in the field. The specific research questions and objectives would also be identified.

The procedure involves running simulations on an RVE consisting of 50 grains and simulating the behaviour of the aluminium using the software Damask. The resulting dataset would include the initial orientations and deformations of each grain, as well as the stresses produced due to these deformations. This data would be arranged into arrays and fed into the convolutional layers of a neural network.

To ensure the accuracy and reliability of the results, appropriate examination methods such as statistical analysis and machine learning techniques would be selected. Cross-validation tests would also be performed to validate the performance of the model and ensure that it is not overfitting the data.

In addition, the project aims to develop a more accurate and efficient method for modelling the anisotropic behaviour of aluminium using neural networks. This would involve careful data handling, feature selection, and target identification to ensure that the model accurately captures the complex behaviour of the material.

4. Work plan, necessary resources

Task	Start Date	End Date	Duration
Gather information	04/15/2023	05/05/2023	21 days
Data handling and pre-processing	05/06/2023	05/20/2023	15 days
Develop neural network model	05/21/2023	06/20/2023	30 days
Train and validate model	06/21/2023	07/21/2023	31 days
Analyse results and draw conclusions	07/22/2023	08/10/2023	20 days
Write report and prepare presentation	08/11/2023	09/10/2023	31 days
Finalize project and submit	09/11/2023	09/30/2023	20 days

To test the statement that neural networks can effectively model the anisotropic behaviour of aluminium, the following methods will be used:

- Statistical analysis: Statistical measures such as mean squared error, R-squared, and correlation coefficients and also perform cross validation on the dataset to evaluate the model.

- Sensitivity analysis: Sensitivity analysis will be performed to understand the effect of different features and hyperparameters on the model's performance.
- Visualization: Visualization tools will be used to analyze the model's predictions and compare them with the actual data.

These methods are especially suitable because they provide a comprehensive and rigorous evaluation of the model's performance and help to identify potential limitations and areas for improvement.

In terms of resources, access to the aluminium dataset generated by the RVE simulations, as well as computational resources for running the neural network model and performing data analysis will be needed. Additionally, specialized software tools for data pre-processing, model development, and visualization may be required. Finally, access to domain experts and experimental resources to validate the model's predictions and evaluate its practical usefulness may also be needed.

5. Literature

1. Alexander E. Mayer, M. V. (2022). Machine-Learning-Based Model of Elastic—Plastic Deformation of Copper for Application to Shock Wave Problem.
2. Darren C. Pagan, C. R. (2022). Graph neural network modeling of grain-scale anisotropic elastic behavior using simulated and measured microscale data.
3. Minyi Dai, M. F. (2021). Graph neural networks for an accurate and interpretable prediction of the properties of polycrystalline materials.

Content presentation created by:

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Student	First examiner (supervisor)	Second examiner (Center of Key Competencies)